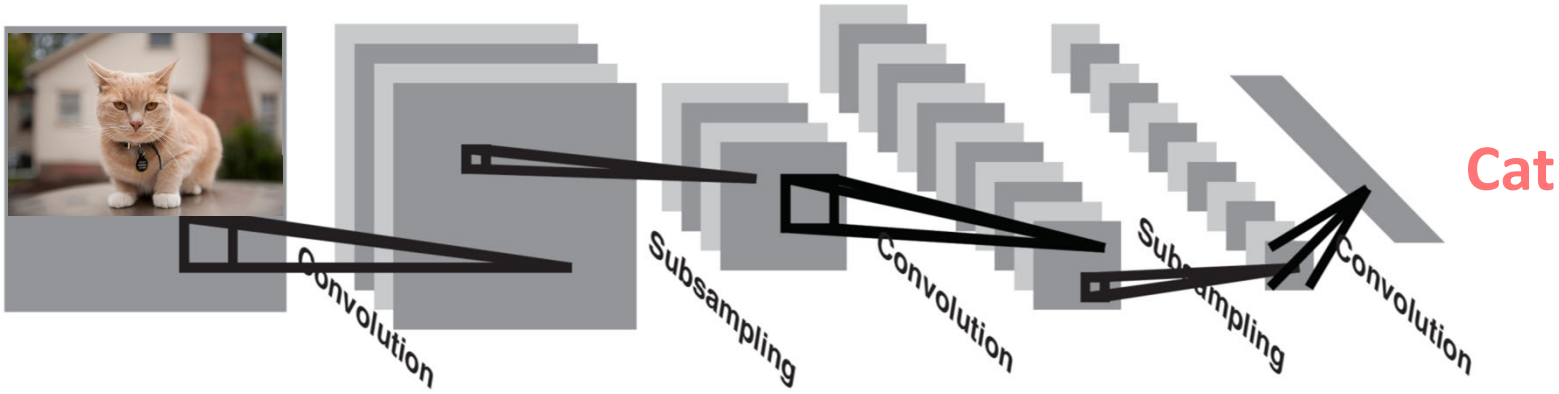
Standard Deep Learning



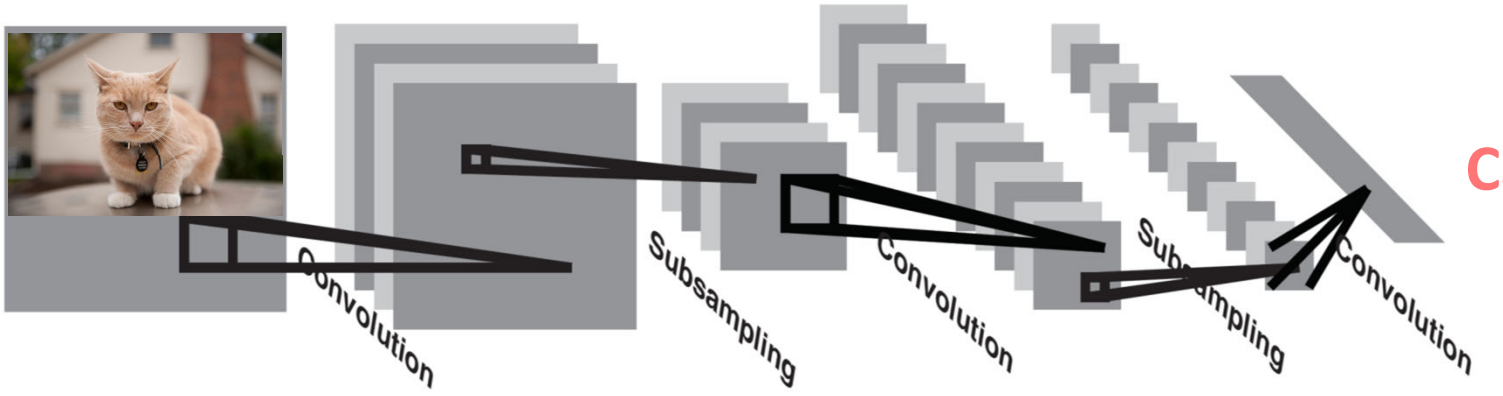
Standard Deep Learning

is discriminative

is a deterministic function

Outputs the predicted label without any accompanying information

Requires fully-labelled data



is opaque

uses feed-forward networks on tensor-like input

Trained and tested in isolation (e.g. from other tasks) -- specialized AI.

Beyond Standard Deep Learning

Standard deep (regression) networks give a point estimate prediction

$$\hat{y} = \underset{y}{\operatorname{argmax}}\, P(y|\mathbf{x})$$

→ With Uncertainty Estimation we try to have an understanding of $P(y|\mathbf{x})$

Beyond mere predictions!



Beyond Standard Deep Learning

Standard deep networks are a point estimate of the parameter distribution:

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, P(\boldsymbol{\theta}|\mathcal{D})$$

→ With probabilistic deep networks we try to have an understanding of $P(\boldsymbol{\theta}|\mathcal{D})$

Beyond deterministic networks



Beyond Standard Deep Learning

In fact the reason we want to understand $P(\boldsymbol{\theta}|\mathcal{D})$ is mainly to be able to understand $P(y|\mathbf{x})$

